# iLearn Practice Exam

12BR · Chapter / Topic 15

## Cambridge A Level Computer Science 9618

Paper 3  Advanced Theory

# Topic 15: Hardware

**Name:** _____

**Class:** _____

**Date:** _____

**Time allowed:** 1 hour 30 minutes

**TOTAL MARKS: 62**

Marks obtained: ..... / 62

**INSTRUCTIONS**

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

**INFORMATION**

· The total mark for this paper is 62.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 This truth table represents a logic circuit.

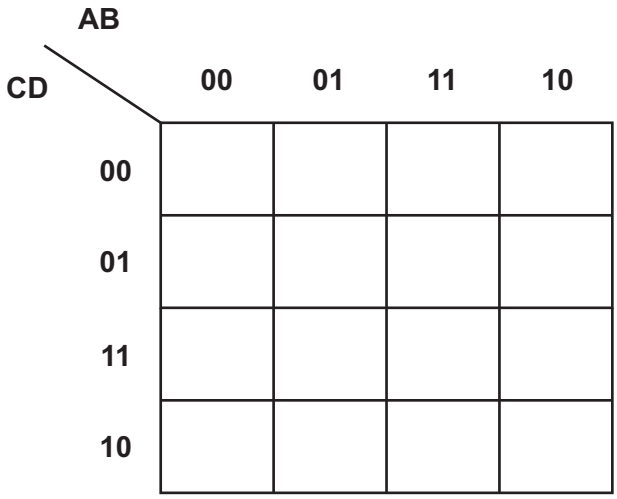
| INPUT | | | | OUTPUT |
|---|---|---|---|---|
| **A** | **B** | **C** | **D** | **Z** |
| 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

**(a)** Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

**Z =** .....

..... [2]

**(b) (i)** Complete the Karnaugh map (K-map) for the given truth table.




[2]

**(ii)** Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

**(iii)** Write the Boolean logic expression from your answer to part **(b)(ii)** as the simplified sum-of-products.

**Z =** .....

..... [2]

2 **(a)** Describe the characteristics of massively parallel computers.

.....

.....

.....

.....

.....

..... [3]

**(b)** State **one** difference and **one** similarity between the SIMD and MISD computer architectures.

difference .....

.....

.....

similarity .....

.....

.....

[2]

3 **(a)** The Karnaugh map (K-map) represents a logic circuit with four inputs.

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| **00** | 0 | 1 | 1 | 0 |
| **01** | 0 | 1 | 0 | 0 |
| **11** | 0 | 1 | 0 | 0 |
| **10** | 0 | 1 | 1 | 0 |

**(i)** Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

**(ii)** Write the Boolean expression from your answer to **6(a)(i)** as a simplified sum-of-products. Do **not** carry out any further simplification.

.....

..... [2]

**(iii)** Write the Boolean expression from your answer to **6(a)(ii)** in its simplest form by using Boolean algebra to simplify it.

..... [1]

**(b)** Simplify the following expression using De Morgan's laws and Boolean algebra.

Show all the stages in your simplification.

$\mathbf{X = \overline{A + B + C} + \overline{B + C}}$

.....

.....

.....

.....

.....

..... [3]

4 Explain what is meant by the term virtual machine.

.....

.....

.....

.....

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.....

.....

..... [4]

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5 This truth table represents a logic circuit.

| INPUT | | | | OUTPUT |
|---|---|---|---|---|
| A | B | C | D | Z |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

**(a) (i)** Complete the Karnaugh map (K-map) for the given truth table.

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | | | | |
| 01 | | | | |
| 11 | | | | |
| 10 | | | | |

[2]

**(ii)** Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

**(iii)** Write the Boolean expression from your answer to **7(a)(ii)** as a simplified sum-of-products.

.....

..... [1]

**(b)** Simplify the following expression using De Morgan's laws and Boolean algebra.

Show all the stages in your simplification.

$$X = \overline{\overline{A}.B.C.\overline{D}} + \overline{\overline{A}.D}$$

.....

.....

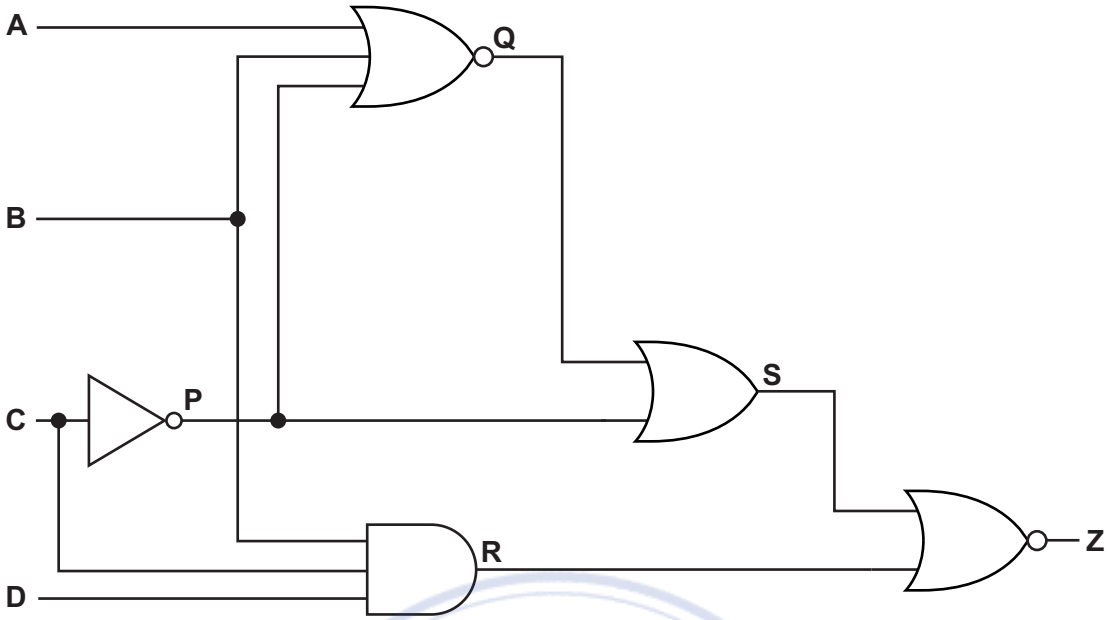
.....

.....

.....

..... [3]

6 The diagram shows a logic circuit.




**(a)** Complete the truth table for the given logic circuit.
Show your working.

| | | | | Working space | | | | |
|---|---|---|---|---|---|---|---|---|
| **A** | **B** | **C** | **D** | **P** | **Q** | **R** | **S** | **Z** |
| 0 | 0 | 0 | 0 | | | | | |
| 0 | 0 | 0 | 1 | | | | | |
| 0 | 0 | 1 | 0 | | | | | |
| 0 | 0 | 1 | 1 | | | | | |
| 0 | 1 | 0 | 0 | | | | | |
| 0 | 1 | 0 | 1 | | | | | |
| 0 | 1 | 1 | 0 | | | | | |
| 0 | 1 | 1 | 1 | | | | | |
| 1 | 0 | 0 | 0 | | | | | |
| 1 | 0 | 0 | 1 | | | | | |
| 1 | 0 | 1 | 0 | | | | | |
| 1 | 0 | 1 | 1 | | | | | |
| 1 | 1 | 0 | 0 | | | | | |
| 1 | 1 | 0 | 1 | | | | | |
| 1 | 1 | 1 | 0 | | | | | |
| 1 | 1 | 1 | 1 | | | | | |

**(b)** Write the Boolean logic expression that corresponds to the given logic circuit as the sum-of-products.

**Z =** .....

..... [1]

**(c)** Use Boolean algebra including De Morgan's laws to simplify the following expression. Show all working.

$$(\overline{\mathbf{A} + \mathbf{B}}) \,.\, (\overline{\mathbf{A} \,.\, \mathbf{B}} + \overline{\mathbf{B} \,.\, \mathbf{C}})$$

Working .....

.....

.....

.....

.....

.....

Simplified expression .....

.....

[4]

7 A virtual machine may be used to emulate computer hardware using software.

Outline **two** benefits and **two** limitations of using a virtual machine.

benefit 1 .....

.....

benefit 2 .....

.....

limitation 1 .....

.....

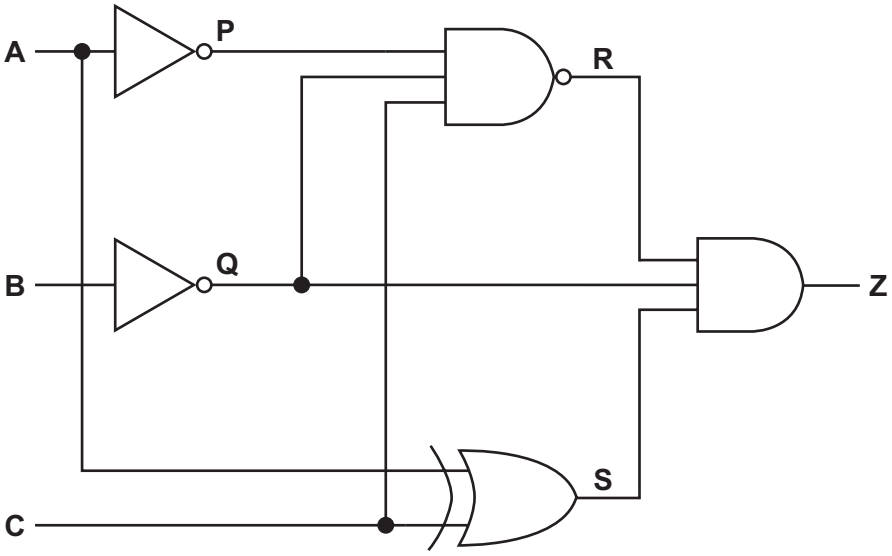
limitation 2 .....

.....

[4]

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8 **(a)** The diagram shows a logic circuit.




Complete the truth table for the given logic circuit.
Show your working.

| | | | working space | | | | |
|---|---|---|---|---|---|---|---|
| **A** | **B** | **C** | **P** | **Q** | **R** | **S** | **Z** |
| 0 | 0 | 0 | | | | | |
| 0 | 0 | 1 | | | | | |
| 0 | 1 | 0 | | | | | |
| 0 | 1 | 1 | | | | | |
| 1 | 0 | 0 | | | | | |
| 1 | 0 | 1 | | | | | |
| 1 | 1 | 0 | | | | | |
| 1 | 1 | 1 | | | | | |

[3]

**(b)** Describe **one** benefit of using a Karnaugh map (K-map) for solving logic problems.

.....

..... [1]

**(c)** The Karnaugh map (K-map) represents a logic circuit with four inputs.

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| **00** | 0 | 0 | 1 | 0 |
| **01** | 0 | 1 | 1 | 0 |
| **11** | 1 | 1 | 1 | 1 |
| **10** | 0 | 0 | 1 | 0 |

**(i)** Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [3]

**(ii)** Write the Boolean expression from your answer to **6(c)(i)** as a simplified sum-of-products. Do **not** carry out any further simplification.

.....

..... [2]

9 This truth table represents a logic circuit.

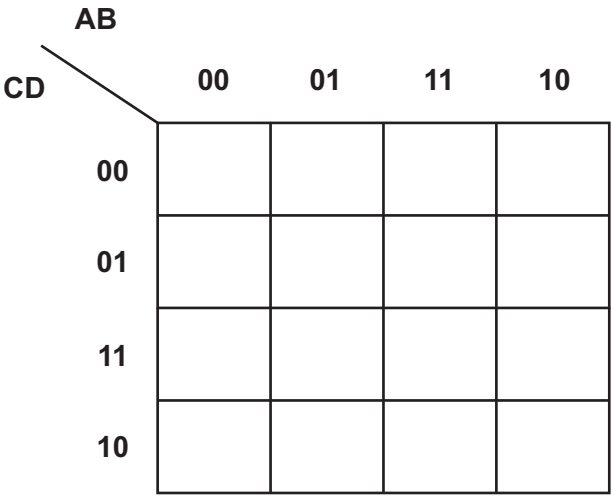
| INPUT | | | | OUTPUT |
|---|---|---|---|---|
| **A** | **B** | **C** | **D** | **Z** |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 |

**(a)** Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

**Z =** .....

..... [2]

**(b) (i)** Complete the Karnaugh map (K-map) for the given truth table.




[2]

**(ii)** Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

**(iii)** Write the Boolean logic expression from your answer to part **(b)(ii)** as the simplified sum-of-products.

**Z =** .....

..... [2]

**END OF QUESTION PAPER**

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